QUANTUMCROP AI – ANALYSIS REPORT

Quantum Remote Sensing & Yield Prediction Platform

Date: 2026-07-26

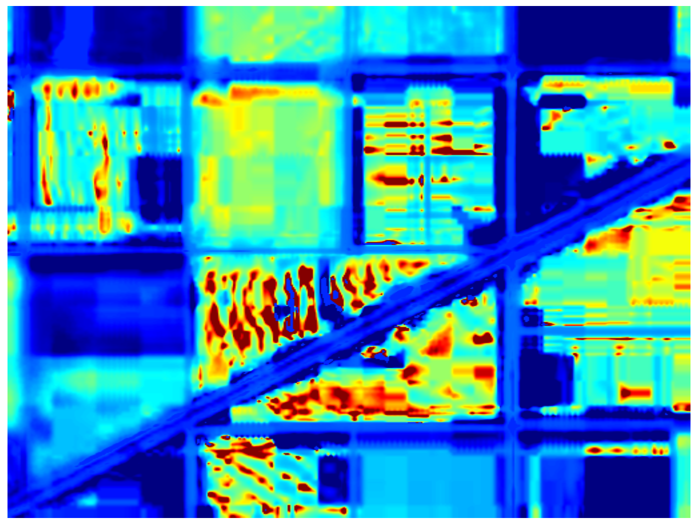
Sensor: Simulated Sentinel-2 Multispectral

Analysis Engine: Classical + QML (Hybrid)

Location ID: Field-Alpha-32

Satellite & Index Imagery

Uploaded Imagery (RGB)



Computed NDVI Index Map

Crop Health & Yield Prediction Summary

Crop Health Status:	**Disease**
Prediction Confidence:	17.81%
Estimated Crop Yield:	**2.74 Tons / Hectare**

Machine Learning Benchmarking (Classical vs Quantum)

Model Name	Type	Health Classification	Confidence	Accuracy (Benchmark)	Inference Latency
RANDOM FOREST	Classical	Severe Stress	55.0%	88.0%	~0.002s
SVM	Classical	Disease	17.8%	86.0%	~0.002s
QSVM	Quantum (QSVM)	Disease	23.2%	84.0%	~0.005s

AI Assistant Recommendations

• Possible Nitrogen Deficiency: Extremely low Chlorophyll Index detected. Apply urea or targeted nitrogen fertilizers.

• High Disease Risk: High spatial variance in NDVI suggests patchy health degradation. Scan for fungal or pest damage.

• Expected Yield Loss: Severe crop stress detected. Crop loss imminent; arrange field check.